

Sugar Slay

Gamified Decision Support System for
Individuals with Type 1 Diabetes



Overview

Sugar Slay Overview

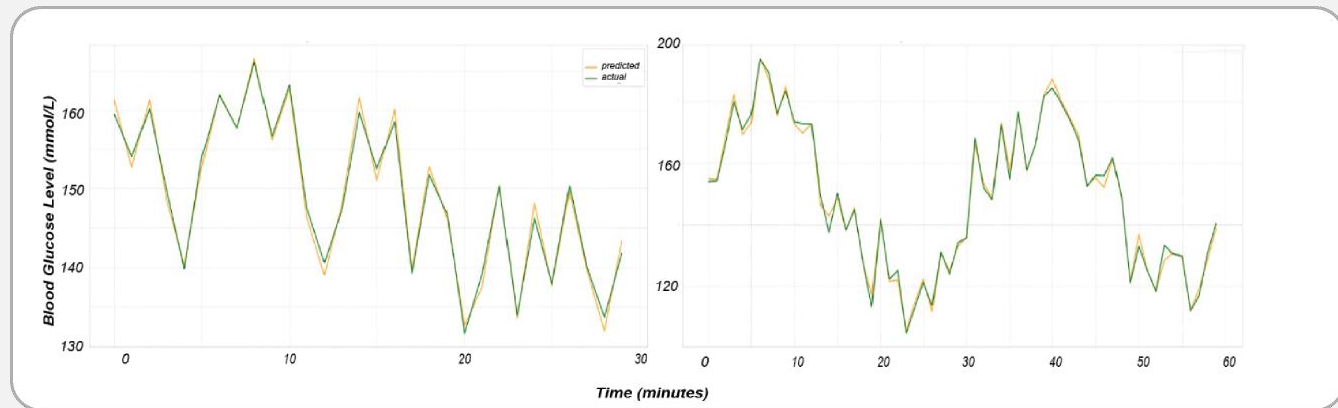
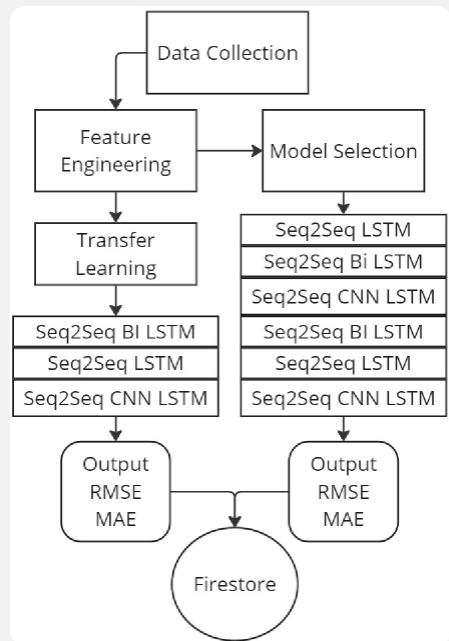
Goals

1. Help users build and sustain healthy habits
2. Encourage self-discovery and exploration
3. Make diabetes management less mundane
4. Improve adolescent experience with T1D

Technology

Sugar Slay Overview

Glucose Trend Prediction



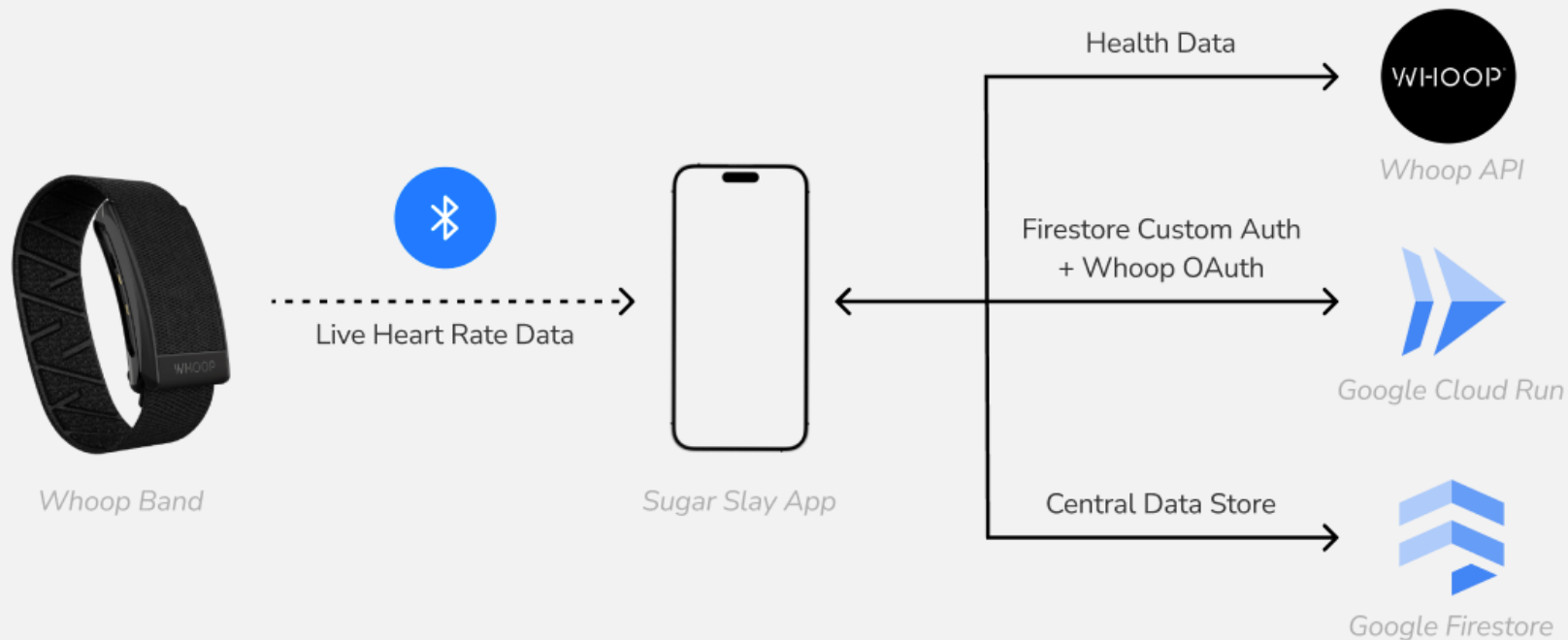
dexcom G6

WHOOOP Integration

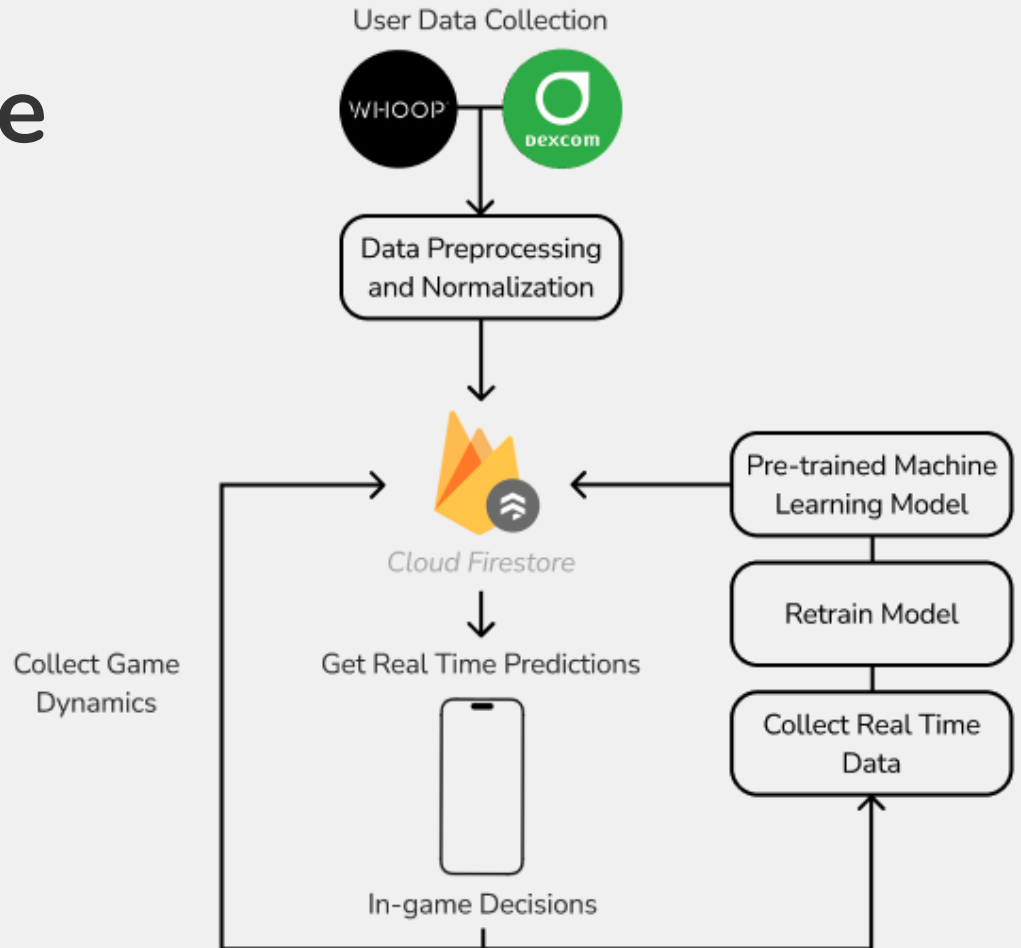
- Allows for the seamless collection of useful health data such as a user's heart rate variability, recovery metrics, sleep quality, and physical activity.
- Always on body, versatile.



Data Architecture

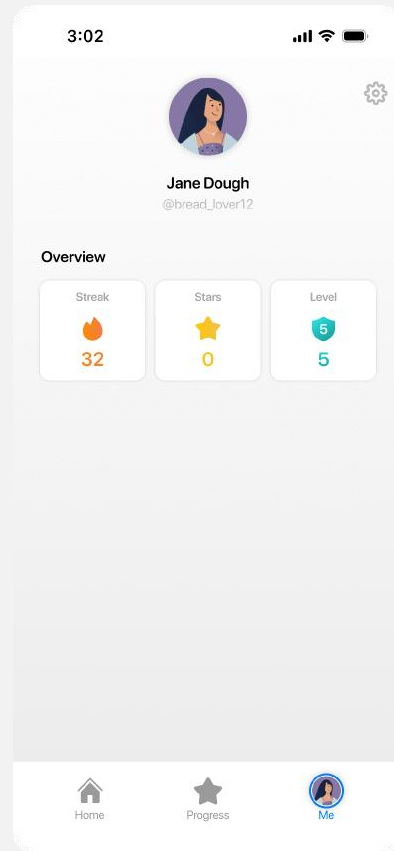
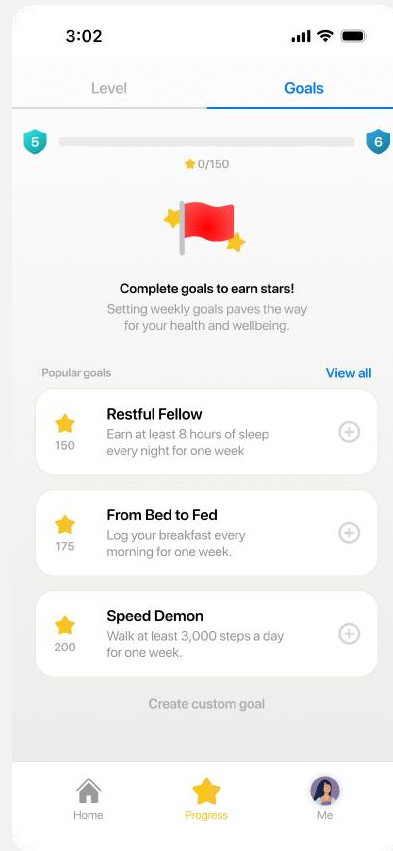
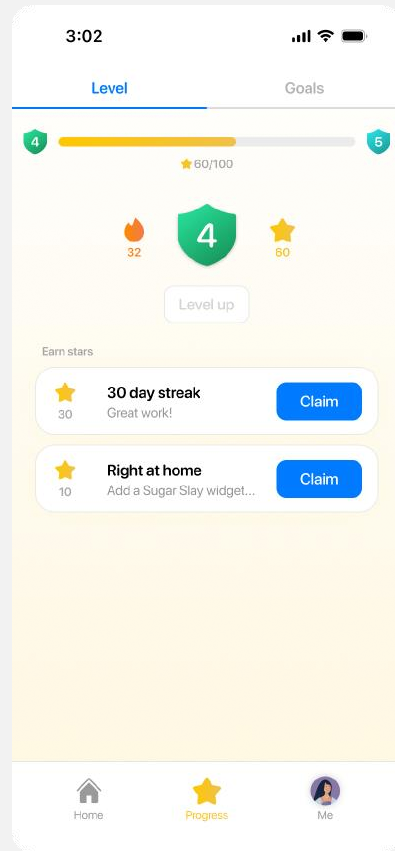
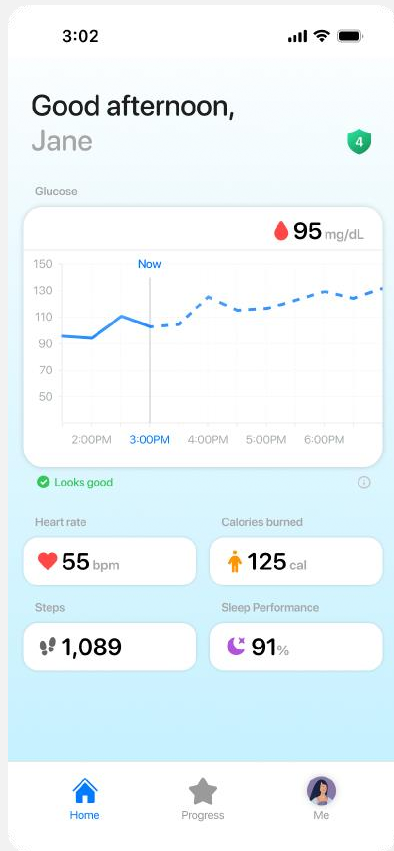


App Architecture



Initial Design

Sugar Slay Overview



Sugar Slay Overview

Feedback on App Design

- Met with Dr. Laura Forlano, CAMD Design professor with Type 1 Diabetes, for feedback.
- Identified need for greater focus on user-centered design and visual refinements.
- Discussed simplifying workflows and improving engagement features before pilot testing.

Design (again)

Refined Design Process

1. Competitor Research
2. User Interviews
3. Developing App Concepts
4. Branding
5. Low Fidelity Prototype
6. Usability Testing
7. High Fidelity Prototype

Competitor Research

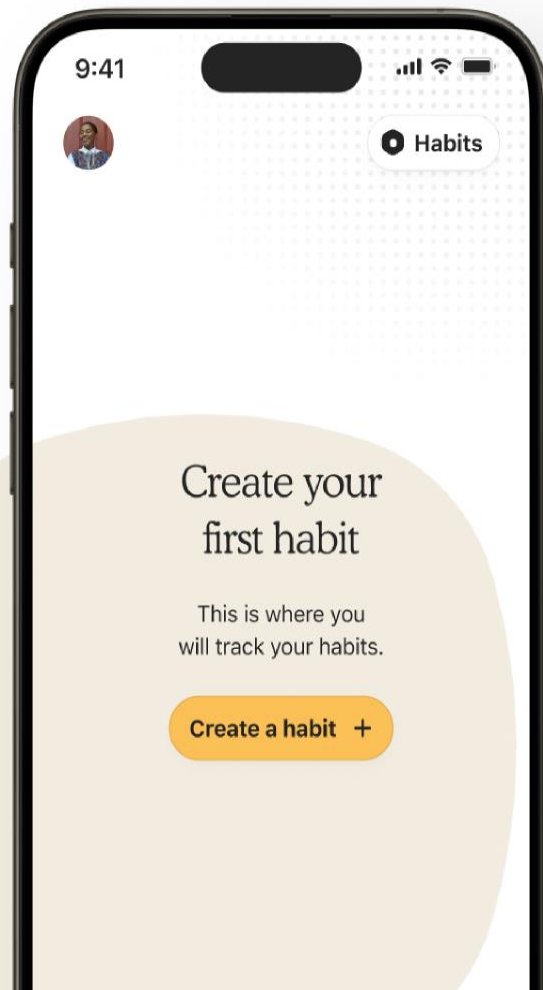
- What are similar apps doing?
- What works, and what doesn't work?

Habit Building Apps

Sugar Slay Overview

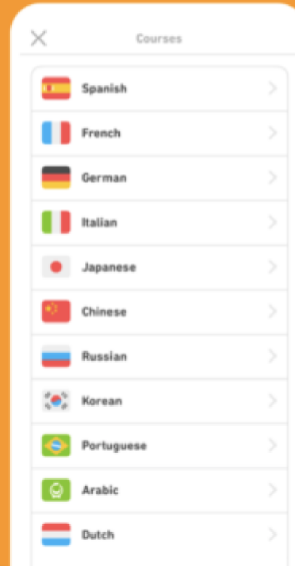
Atoms.

Tiny changes,
remarkable results.

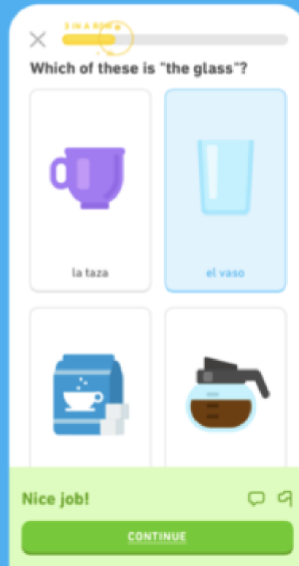


duolingo

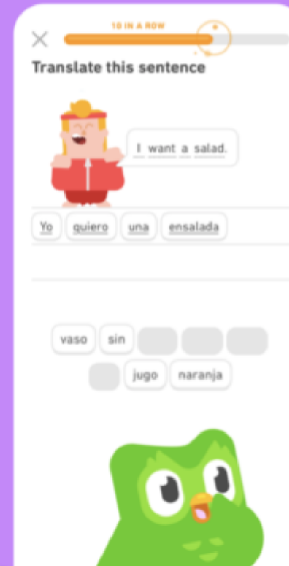
Spanish, French
and 30 more
languages free



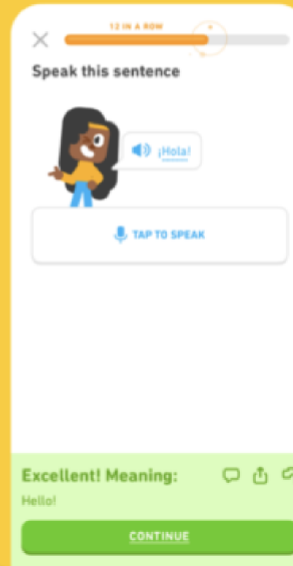
The fastest way
to learn a new
language



Quick, fun
bite-sized lessons
that work



Speak and
hear the
language

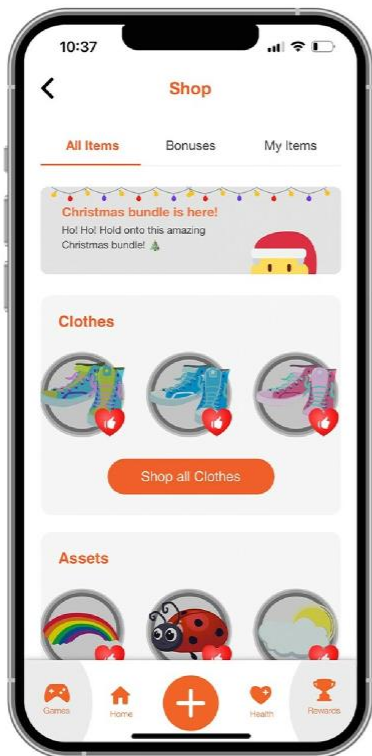


Diabetes Apps

Sugar Slay Overview

Happy Bob





Misc. Wellness Apps



Bearable App

Health tracking for wellbeing, not fitness.



400k+ users

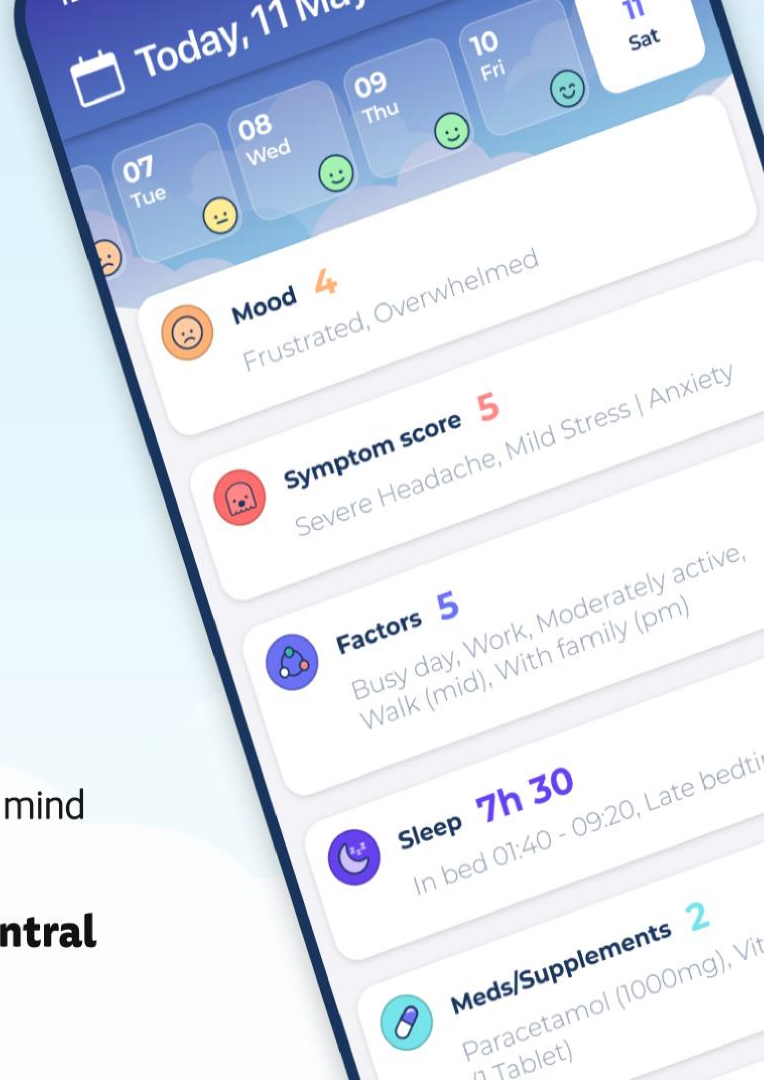


85% 5-Stars



verywellmind

PsychCentral



Competitor research gave us an idea of what *most* users want in an app.

But what do *our* users want?

User Interviews

- What problems are people with T1D facing?
- How might Sugar Slay solve them?

User stories

As a [type of user] I want to [goal or action] so that [desired outcome]

Cole

As a person with diabetes I want to know when my next spike will come so I can ensure that I'll have what I need

Cole

As a new diabetic, I want a catalog of resources so that I can better acclimate to living with diabetes

Cole

As a person with diabetes, I want to know my hard work is recognized so that I don't feel as if I'm going through this alone

Cole

As a person with diabetes, I want to be able to log only the data that applies to me, so that I don't feel overwhelmed with all the types of data I'm being asked to enter

Cole

As a person with diabetes, I want to know when my next spike will be, so that I can be prepared for when it comes

Cole

As a new diabetic, I want to know what options I have to manage my diabetes so that I can best take care of myself

Cole

As a person with diabetes, I want an easy way to check if my glucose is in range while I'm working out so that I can best care for myself

Cole

As a person with diabetes, I want a calmer, more neutral indication of my level trend so that I don't freak out or become overwhelmed

Cole

As a person with diabetes, I want a more fun way to manage my condition so that it feels more satisfying and less monotonous

Cole

As a person with diabetes, I want flexibility in my diabetes management so that when traveling, I don't feel like I'm failing

Cole

As a person with diabetes, I want to feel a sense of community so that I have people to relate to

Cole

As a person with diabetes, I want someone to comfort me and give me guidance so that I don't get too down on myself about having diabetes.

Cole

As a person with diabetes, I want someone to motivate me to eat healthier, to run, to workout, so that I can lead a healthier lifestyle.

Cole

As a person with diabetes, I want an easy way to know how many carbs are in the food I'm eating so that I don't accidentally spike my sugar level

Cole

As a person with diabetes, I want access to other people with diabetes so I can learn from them how to best take care of myself.

Cole

As a person with diabetes, I want to feel a sense of community so that I know I'm not going through this alone

Cole

As a person with diabetes, I want to be able to customize how I track my meals so that I don't over-manage my diet and follow unhealthy eating practices

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As a person with diabetes, I want to visualize what impacts my mood so that I can know what I need to change in my diabetes management

Cole

As a new diabetic, I want the support of my friends and family so that I feel comfortable as I get used to having diabetes

Cole

As a person with diabetes, I want to receive fewer notifications so that I feel less annoyed by my devices

Cole

As a person with diabetes, I want to celebrate diabetes awareness month so that I can eat some sweets and acknowledge how hard I've worked to get here

Cole

As a person with diabetes, I want to be reminded to stock up on insulin/meds so that I don't run out and end up unable to care for myself

Cole

As a person with diabetes, I want to be able to take a day off when my blood sugar is low so that I can just focus on resting

Cole

As a person with diabetes, I want to celebrate my Diaversary so that I can eat some sweets and acknowledge how hard I've worked in the past

Cole

As a teen with diabetes, I want a way to manage the alerts my parents receive so that they don't bombard me with texts when my levels are slightly lower than normal

Cole

As a person with diabetes, I want to share my progress with my friends and family so that they can see how hard I work

Cole

As a person with diabetes, I want to give my device a nickname so that I feel like I have more ownership over my diabetes management

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As a person with diabetes, I want to feel like I'm being taken care of so that I don't feel so alone in managing my health

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Users want to feel supported

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Users want to be rewarded

Sugar Slay Overview

As a person with diabetes,
I want to share my
progress with my friends
and family so that they
can see how hard I work

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As a person with diabetes, I
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Users want to belong

As a person with diabetes,
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As a person with diabetes,
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Developing App Concepts

- What core features will power Sugar Slay and make it an app that people will use?

User stories

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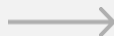
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Diaversary

- Congratulatory message
- Option to share to Instagram / other social platforms

Cole

Blood Sugar Data Visualization

- See what trend looks like
- Understand when my next spike might be (and how likely it is)

Cole

Resources / Wisdom Page

- Guides and How-tos
- Doctor co-signed articles

Cole

Streaks

- TBD

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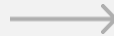
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Notifications/Alerts

- Warns user if blood sugar is about to spike
- Option to change how aggressive the alerts are

Cole

Virtual Pet/Plant

- Might act as something like a nurse/patient combo
- TBD

Cole

Community Page

- Allows users to connect and share tips/memes

Cole

Customized Food Log

- Allows user to set what types of dietary data, if any, they want to log

Cole

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Mood insight

- Helps user understand what actions influence their mood

Cole

Mood logging

- Allows the user to log their mood and add a note

Cole

Parent/Child Shared settings

- Allows child and parent to agree on what the parent will see (re: notifications, data)

Cole

Levels

- Motivates user to continue caring for their diabetes.
- User can share their progress

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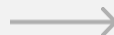
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Cole



Device nickname

- User can label their devices
- Helps users gain a sense of ownership over their diabetes management

Cole

More customized logging

- Allows the user to select what types of data they log.
- One tab for logging data. Distinct process for logging each type of data.

Cole

Sugar forecast

- Visual weather-report like feature to view when next spike will be (with likelihood)

Cole

Carb "checker" (no logging)

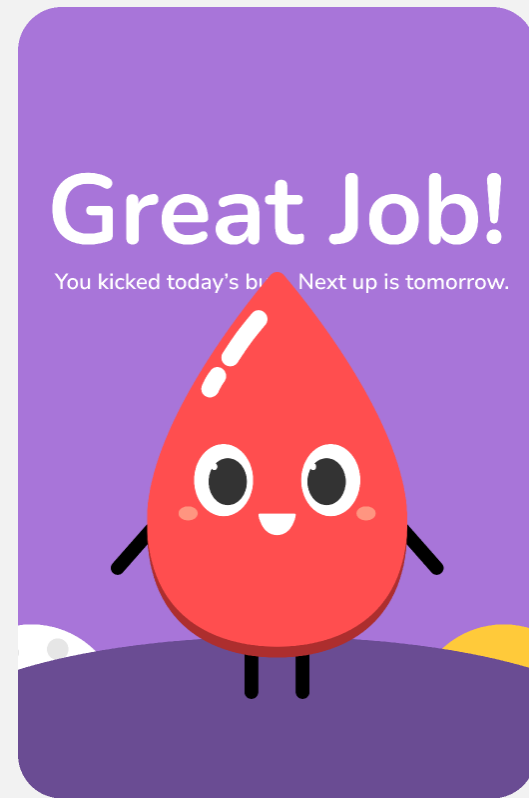
- Way for the user to scan a food item and see whether or not they should eat it (depending on their sugar level)
- Doesn't show carb/caloric details automatically. User needs to click to view possibly triggering details.

Cole

Features

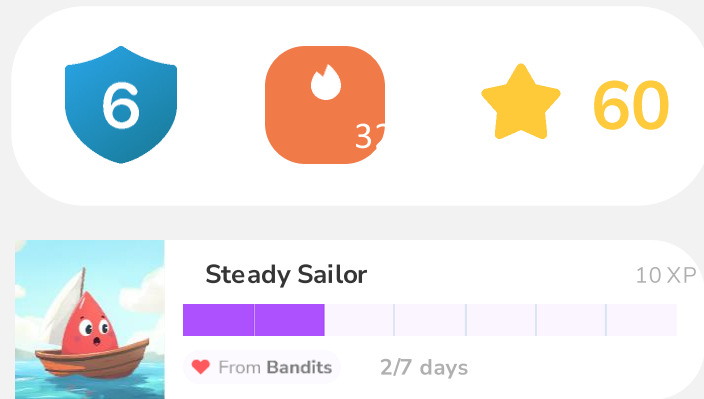
Users want to feel supported

- Motivational rhetoric
- A comforting persona
- Personalized messages



Users want to be rewarded

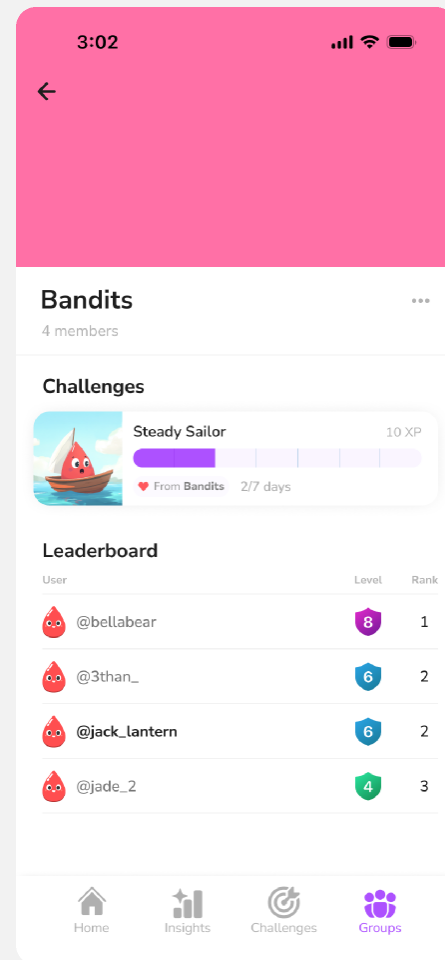
- Streaks, Challenges, Badges, and other gamified features
- Haptic feedback
- Ability to share progress



Users want to belong

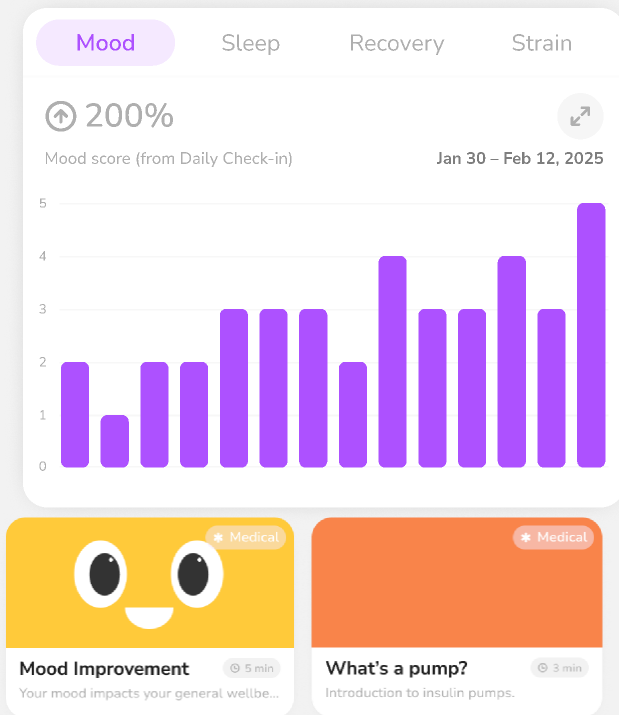
- Community features
- Leaderboards
- Family features

Sugar Slay Overview



Users want to learn

- Personalized health insights
- Intelligent recommendations
- Diabetes resources



Branding

- How should users perceive Sugar Slay?
- What should the app look/feel like?



Playful, Motivational, Curious, Warm, Inclusive

Core Colors

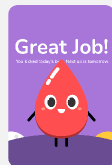
These four colors drive Sugar Slay. They appear all throughout the app and the user will associate them heavily with their diabetes care.



3

Secondary Colors

These colors will be sprinkled in tiny doses all throughout the app, but will take center stage in illustrations and animations.



5

Bluddy's Palette

The colors of our favorite bluddy friend. They're confident and friendly and provokes a little mischief.



4

Nunito

Nunito is used for titles, headers, and any other large text that features big, bold language.

- n1 **Keep it up!**
- n2 **You're on a roll. Let's go!**
- n3 **Featured communities**

6

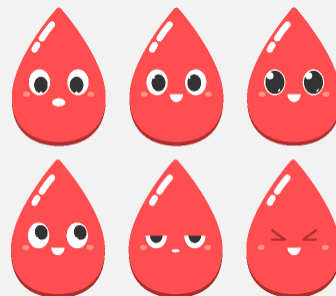
Nunito Sans

Nunito Sans is used for body text and buttons.

- p1 I'm feeling okay right now. I'm hoping I'll feel better soon though.
- p2 Are you sure? Sounds good.
- p3 Account

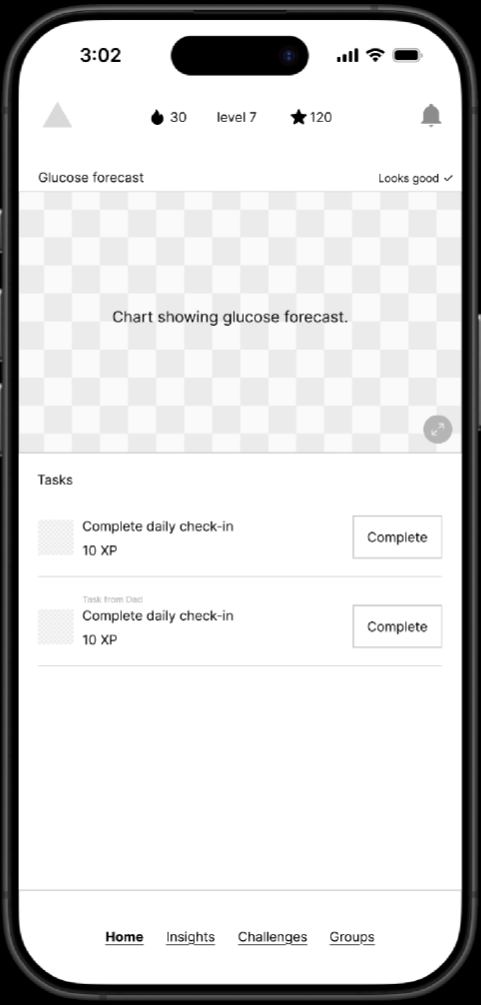


7



Low Fidelity Prototype

- Basic wireframe of the app



3:02



30

level 7



120



Glucose forecast

Looks good ✓

Chart showing glucose forecast.



Tasks



Complete daily check-in
10 XP

Complete

Task from Dad



Complete daily check-in
10 XP

Complete

[Home](#) [Insights](#) [Challenges](#) [Groups](#)

Usability Testing

- What needs to be changed about the design before we continue to work on it?

Usability Testing

- Purpose
 - Evaluate ease of use, engagement, and clarity of key features (onboarding, logging, challenges, insights).
 - Identify pain points and user-driven improvement ideas.
- Participants
 - 7 adolescents (13–18) with Type 1 Diabetes
 - Mix of newly diagnosed and long-term users

Key Insights

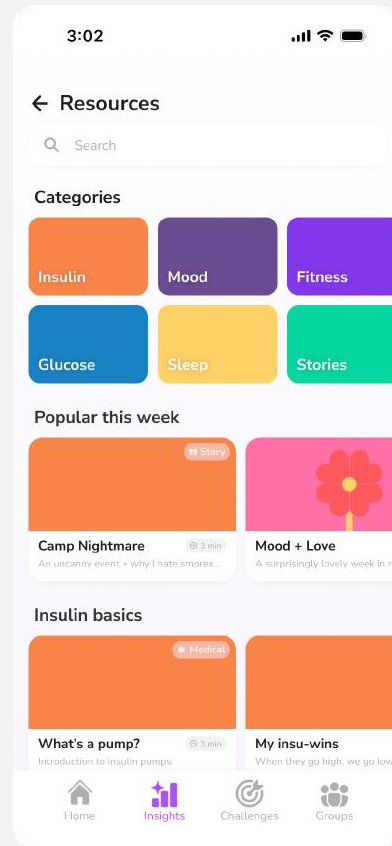
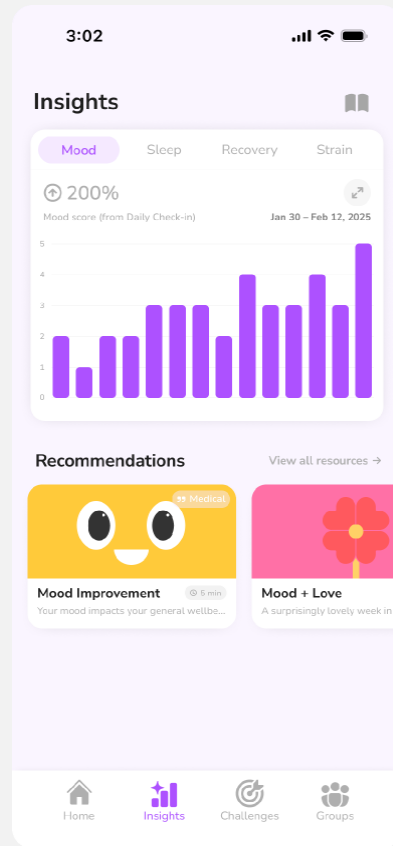
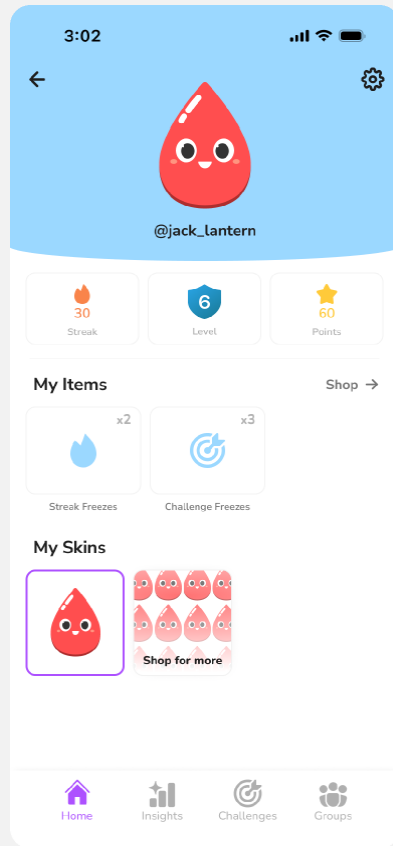
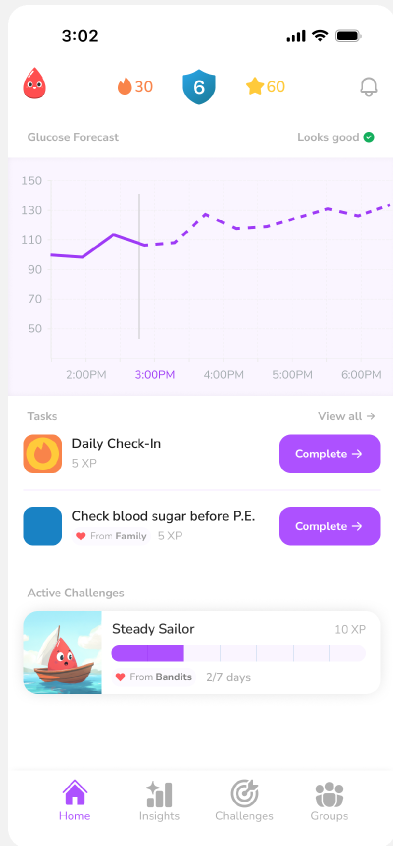
- **Onboarding:** Clear explanations but requests for stronger visual cues and notes/journal options
- **Logging:** Meal logging easy; insulin logging confusing (bolus vs. short-acting)
- **Gamification & Challenges:** Motivating but desire for progressive unlocking and health context
- **Glucose Forecast:** Viewed as unique and helpful; clearer range presentation suggested
- **Device Integration:** High interest in Dexcom/WHOOOP connections for real-time insights

Final design phase

High Fidelity Prototype

- Transforming the lo-fi prototype into a beautiful, highly detailed mockup that reflects the final look of the app.





Sugar Slay Overview

Development

- Transforming the hi-fi prototype into code.



10:45



Sugar Slay

Diabetes management, made fun!

Get started

I already have an account

Upcoming Milestones

- What's next for Sugar Slay?



- **Companion App**
 - Create a lightweight companion app for caregivers/parents to view trends, alerts, and progress
 - Enables supportive oversight without overwhelming users
- **Model Personalization & Integration**
 - Adapt Seq2Seq BiLSTM model to individual CGM and WHOOP data
 - Integrate real-time streams for personalized glucose forecasts
- **Clinical Pilot Study (First Step)**
 - Collect real-time data (CGM + WHOOP) within app
 - Measure Time in Range (TIR) and engagement as primary outcomes

- **Expanded User Studies**
 - Conduct larger usability tests with diverse adolescent populations
 - Validate gamification features and engagement strategies at scale
- **Feature Enhancements**
 - Refine challenge progression, visuals, and insights clarity
 - Develop SlayBot with strict safety guardrails
- **Long-Term Engagement & Scaling**
 - Analyze retention and adherence over extended use
 - Prepare for broader deployment and expansion

Thank you

